

MediScan AI - Project Description

MediScan AI — Multi-Modal Medical Imaging Diagnostic Assistant

Overview

MediScan AI is a multi-modal deep learning system that screens three of the most common and clinically urgent diagnostic scenarios in radiology — **brain tumors**, **pneumonia**, and **COVID-19** — from a single web interface. Three separate convolutional neural networks, each built on an EfficientNet backbone pretrained on ImageNet, were trained using two-phase transfer learning (frozen head-training followed by careful low-learning-rate fine-tuning) to classify Brain MRI, Chest X-Ray, and Chest CT images respectively.

The three models were then unified into a single production-ready application built with **Gradio** and deployed on **Hugging Face Spaces**. Users upload or photograph a scan, provide basic patient details, and receive an AI-generated diagnosis with a confidence score, a color-coded severity rating, and plain-language guidance on recommended next steps. A built-in AI chatbot, powered by Groq’s `llama-3.3-70b-versatile`, is automatically aware of the user’s latest scan result and can answer follow-up questions about the diagnosed condition — while staying strictly within its three supported medical topics.

Across the three tasks, the models reached **95.6% accuracy / 0.99 AUC** (brain tumor, 4-class), **91.3% accuracy / 0.97 AUC** (COVID-19 CT), and **89-92% accuracy / 0.95 AUC** (pneumonia X-ray), with special attention paid throughout to recall — the metric that matters most when a missed diagnosis carries real clinical risk.

Key Features

- Three independently trained, disease-specific deep learning classifiers unified in one app
- Two-phase transfer learning (frozen backbone → targeted fine-tuning) for each model
- Confidence-aware UI with severity color-coding, low-confidence warnings, and a probability-breakdown donut chart
- Context-aware AI chatbot that references the user’s most recent diagnosis
- Session-based patient scan history log
- Upload, webcam capture, or clipboard-paste image input; light/dark theme toggle
- Clear medical disclaimers throughout — built as an educational/research screening tool, not a diagnostic device

Technologies Used

Python · TensorFlow / Keras · EfficientNetB0 / B3 / B4 · scikit-learn · NumPy · Pandas · Matplotlib · Seaborn · Gradio · Groq API (Llama 3.3 70B) · Kaggle / kagglehub · Hugging Face Spaces

Machine Learning Models

Model	Backbone	Task	Input Size
Brain Tumor Classifier	EfficientNetB0	4-class MRI classification (glioma, meningioma, pituitary tumor, no tumor)	224×224
COVID-19 CT Classifier	EfficientNetB4	Binary CT classification (COVID vs. non-COVID)	380×380
Pneumonia X-Ray Classifier	EfficientNetB3	Binary X-ray classification (pneumonia vs. normal)	384×384

Dataset

- **Brain Tumor MRI Dataset** (Kaggle) — 7,200 balanced MRI images across 4 classes
- **SARS-CoV-2 CT-Scan Dataset** (Kaggle) — 2,481 chest CT images, COVID vs. non-COVID
- **Chest X-Ray Images (Pneumonia)** (Kaggle, Paul Mooney) — 6,464 chest X-ray images, pneumonia vs. normal

Results Summary

Model	Accuracy	ROC-AUC	F1-Score (macro)
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Brain Tumor (EfficientNetB0)	95.6%	0.99	0.95
COVID-19 CT (EfficientNetB4)	91.3%	0.97	0.91
Pneumonia X-Ray (EfficientNetB3)	89-92%	0.95-0.96	0.89-0.93

Skills Demonstrated

- Transfer learning and staged fine-tuning strategy design for CNNs on medical imaging data
- Handling class imbalance with computed class weights, and mitigating catastrophic forgetting during fine-tuning
- Rigorous model evaluation: confusion matrices, per-class precision/recall/F1, ROC-AUC, and custom composite scoring
- Targeted error analysis and iterative model improvement (identifying and fixing a specific false-negative pattern)
- End-to-end ML product development: from raw Kaggle datasets to a deployed, multi-model, LLM-augmented web application
- Debugging real production issues (probability/logits normalization, class-index ambiguity) in a live deployment
- Full-stack AI app development with Gradio, including custom UI/UX, theming, and an integrated conversational assistant